

Question 1.1 An ad for ADT Security Systems says,

“When you go on vacation, burglars go to work [...] According to FBI statistics, over 25% of home burglaries occur between Memorial Day to Labor Day.”

Do the data in the ad support the claim that burglars are more likely to go to work during the time between Memorial Day to Labor Day? Please explain your answer. **(6 Points)**

Note: You can assume that “over 25%” means only slightly over. Had it been much over, say closer to 30%, then the marketers would have said so.

Note: Memorial Day is observed on the last Monday of May and Labor Day is observed on the first Monday of September.

The data do not support the claim that burglars go to work between Memorial Day and Labor Day. Memorial Day occurs late in May and Labor Day occurs early in September, meaning that there are about 3 months in between the 2 holidays, which equates to about 25% of the calendar year. The data says that 25% of home burglaries occur during this time, suggesting that most home burglaries occur at a constant rate throughout the entire calendar year.

Question 5.1. The data were gathered by the following procedure, reported in the study. “Between January and June 1998, parents of children aged 2-16 years [...] that were seen as outpatients in a university pediatric ophthalmology clinic completed a questionnaire on the child’s light exposure both at present and before the age of 2 years.” Was this study observational, or was it a controlled experiment? Explain. **(5 Points)**

This was an observational study. In a controlled experiment, there are manipulated variables or treatment that may be assigned to groups of individuals. But, in this study, data is being collected without any variables being manipulated.

Question 5.2. The study found that of the children who slept with a room light on before the age of 2, 55% were myopic. Of the children who slept with a night light on before the age of 2, 34% were myopic. Of the children who slept in the dark before the age of 2, 10% were myopic. The study concluded the following: “The prevalence of myopia [...] during childhood was strongly associated with ambient light exposure during sleep at night in the first two years after birth.”

Do the data support this statement? Why or why not? You may interpret “strongly” in any reasonable qualitative way. **(5 Points)**

The data does support the statement. An association is any relation or link. According to the study, in the rooms that had less light exposure, less children were myopic. While other factors may be at play, there is a definite, strong association between light exposure and the percentage of myopic children.

Question 5.3. On May 13, 1999, CNN reported the results of this study under the headline, “Night light may lead to nearsightedness.” Does the original study claim that night light causes nearsightedness? (5 Points)

The original study does not claim that night light causes nearsightedness. The connection between light exposure and percentage of myopic children is merely an association. Additionally, the claim from the report that night light causes nearsightedness is not entirely correct, as there was a higher percentage of myopic children who slept with a room light on than that of myopic children who slept with a night light on.

Question 5.4. The final paragraph of the CNN report said that “several eye specialists” had pointed out that the study should have accounted for heredity.

Myopia is passed down from parents to children. Myopic parents are more likely to have myopic children, and may also be more likely to leave lights on habitually (since the parents have poor vision). In what way does the knowledge of this possible genetic link affect how we interpret the data from the study? Explain. **(5 Points)**

The genetic link means that there is no causal relationship that can be found within the original study, and that only an association can be found. Given this new information that was previously not accounted for, a new study would have to be done involving the children's parents and whether or not they also have myopia.

